

Hospital Data Analysis Using SQL

Project Documentation

Project Title: Hospital Data Analysis Using SQL

Prepared By: Varsha Maurya

Role: Aspiring Data Analyst

Tools Used: MySQL Workbench, SQL, Microsoft Word

Database: MySQL

Table Name: hospital_data

Dataset: Hospital_Data.csv

1. Introduction

Hospitals generate a significant amount of operational data every day, including patient admissions, doctor availability, hospital departments, discharge details, and medical expenses. Analyzing this data helps healthcare organizations improve resource allocation, monitor patient trends, evaluate departmental performance, and manage operational costs.

This project demonstrates how SQL can be used to analyze hospital data by answering business-oriented questions using aggregate functions, grouping, sorting, and date functions.

2. Project Objectives

The objectives of this project are:

- Calculate the total number of patients across hospitals.
- Analyze the average number of doctors available in each hospital.
- Identify departments with the highest and lowest patient counts.
- Determine the hospital with the highest medical expenses.
- Calculate daily average medical expenses.
- Analyze patient hospital stay durations.
- Generate city-wise patient reports.
- Generate department-wise average stay reports.
- Generate monthly medical expense reports.
- Practice SQL queries used in real-world healthcare data analysis.

3. Tools and Technologies

Tool	Purpose
MySQL Workbench	Database Management
SQL	Data Querying and Analysis
Microsoft Word	Documentation
PDF	Final Project Submission

4. Dataset Description

The Hospital_Data.csv dataset contains hospital-related information used for SQL analysis.

Column Name	Description
Hospital Name	Name of the hospital
Location	City where the hospital is located
Department	Medical department
Doctors Count	Number of doctors available
Patients Count	Number of patients treated
Admission Date	Date of patient admission
Discharge Date	Date of patient discharge
Medical Expenses	Total medical expenses

5. Database Creation

```
CREATE DATABASE hospital_analysis;
```

```
USE hospital_analysis;
```

6. Importing Dataset

The Hospital_Data.csv file was imported into MySQL Workbench using the **Table Data Import Wizard**. After successful import, the dataset was stored in the **hospital_data** table for analysis.

7. SQL Analysis

Query 1: Total Number of Patients

Objective

Calculate the total number of patients across all hospitals.

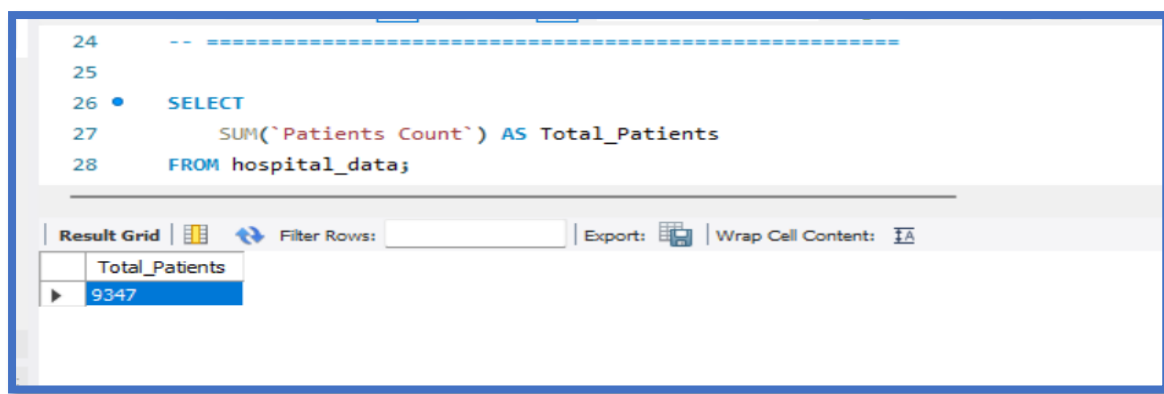
SQL Query

```
SELECT
```

```
    SUM(`Patients Count`) AS Total_Patients
```

```
FROM hospital_data;
```

Output



The screenshot shows a SQL query editor with the following code:

```
24  -- =====
25
26  •  SELECT
27      SUM(`Patients Count`) AS Total_Patients
28  FROM hospital_data;
```

Below the editor is a toolbar with options: Result Grid, Filter Rows, Export, and Wrap Cell Content. The Result Grid shows the output of the query:

Total_Patients
9347

Observation

The query calculates the total number of patients by summing all values in the **Patients Count** column.

Query 2: Average Number of Doctors per Hospital

Objective

Calculate the average number of doctors available in each hospital.

SQL Query

```
SELECT
```

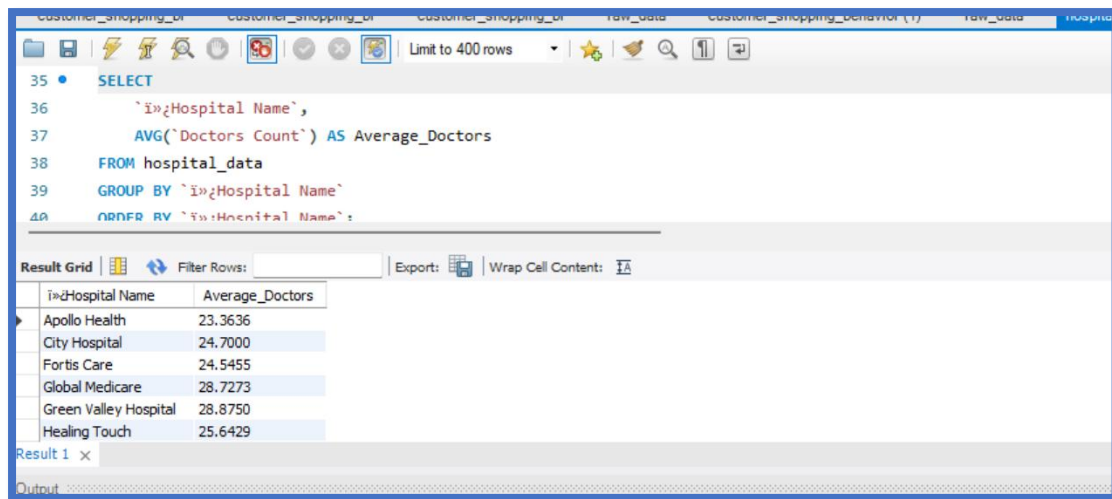
```
    `Hospital Name`,
```

```
    AVG(`Doctors Count`) AS Average_Doctors
```

```
FROM hospital_data
```

```
GROUP BY `Hospital Name`;
```

Output



```

35 SELECT
36     `Hospital Name`,
37     AVG(`Doctors Count`) AS Average_Doctors
38 FROM hospital_data
39 GROUP BY `Hospital Name`
40 ORDER BY `Hospital Name`

```

Hospital Name	Average_Doctors
Apollo Health	23.3636
City Hospital	24.7000
Fortis Care	24.5455
Global Medicare	28.7273
Green Valley Hospital	28.8750
Healing Touch	25.6429

Observation

The query groups the records by hospital and calculates the average number of doctors available in each hospital.

Query 3: Top 3 Departments with Highest Number of Patients

Objective

Identify the three departments treating the highest number of patients.

SQL Query

SELECT

Department,

SUM(`Patients Count`) AS Total_Patients

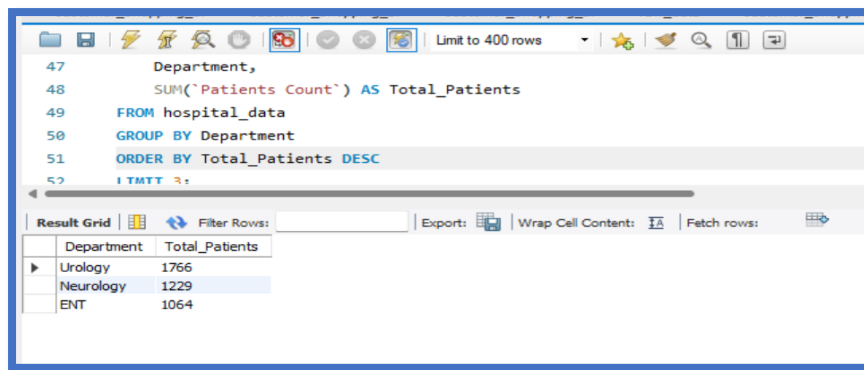
FROM hospital_data

GROUP BY Department

ORDER BY Total_Patients DESC

LIMIT 3;

Output



```

47     Department,
48     SUM('Patients Count') AS Total_Patients
49 FROM hospital_data
50 GROUP BY Department
51 ORDER BY Total_Patients DESC
52 LIMIT 3;

```

Department	Total_Patients
Urology	1766
Neurology	1229
ENT	1064

Observation

The query identifies the top three departments based on the total number of patients treated.

Query 4: Hospital with Maximum Medical Expenses

Objective

Find the hospital with the highest total medical expenses.

SQL Query

SELECT

 'Hospital Name',

 SUM('Medical Expenses') AS Total_Medical_Expenses

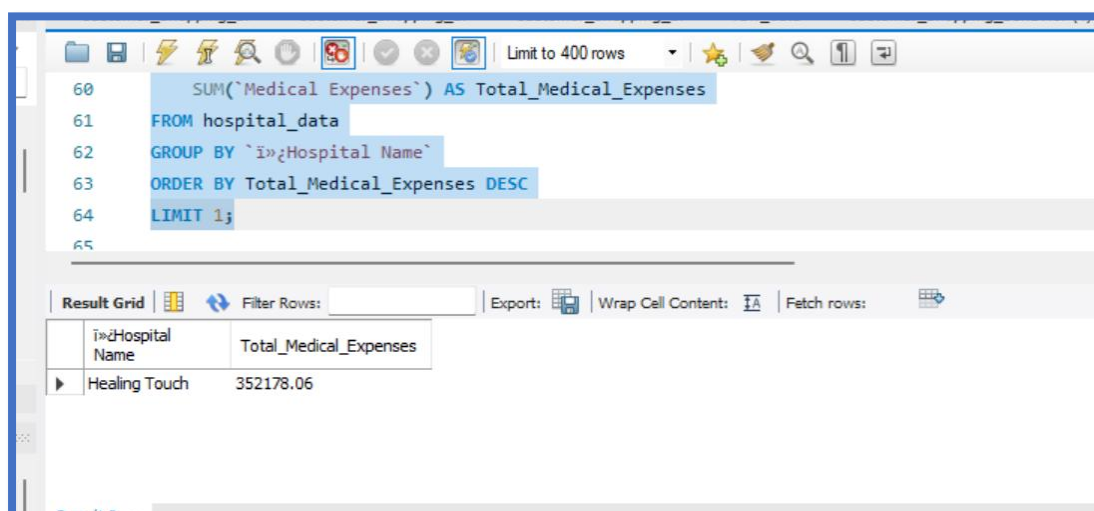
FROM hospital_data

GROUP BY 'Hospital Name'

ORDER BY Total_Medical_Expenses DESC

LIMIT 1;

Output



```

60     SUM('Medical Expenses') AS Total_Medical_Expenses
61 FROM hospital_data
62 GROUP BY 'Hospital Name'
63 ORDER BY Total_Medical_Expenses DESC
64 LIMIT 1;
65

```

'Hospital Name'	Total_Medical_Expenses
Healing Touch	352178.06

Observation

The query identifies the hospital that recorded the highest medical expenses.

Query 5: Daily Average Medical Expenses

Objective

Calculate the average medical expenses per day for each hospital.

SQL Query

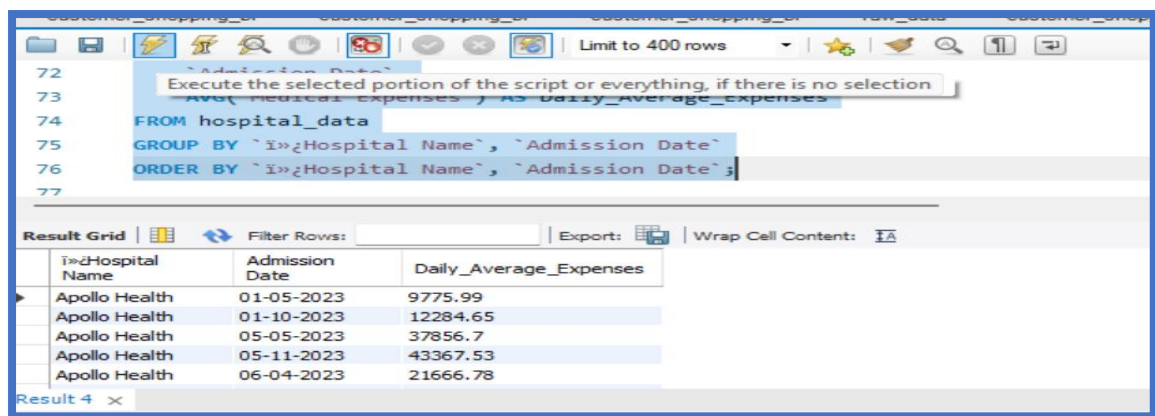
SELECT

```
`Hospital Name`,  
`Admission Date`,  
AVG(`Medical Expenses`) AS Daily_Average_Expenses
```

FROM hospital_data

GROUP BY `Hospital Name`, `Admission Date`;

Output



The screenshot shows a SQL IDE window with a query editor and a results grid. The query editor contains the following SQL code:

```
72 `Admission Date`,  
73 AVG(`Medical Expenses`) AS Daily_Average_Expenses  
74 FROM hospital_data  
75 GROUP BY `Hospital Name`, `Admission Date`  
76 ORDER BY `Hospital Name`, `Admission Date`;  
77
```

The results grid displays the following data:

Hospital Name	Admission Date	Daily_Average_Expenses
Apollo Health	01-05-2023	9775.99
Apollo Health	01-10-2023	12284.65
Apollo Health	05-05-2023	37856.7
Apollo Health	05-11-2023	43367.53
Apollo Health	06-04-2023	21666.78

Observation

The query calculates the average medical expenses for each hospital on every admission date.

Query 6: Longest Hospital Stay

Objective

Find the record with the longest hospital stay.

SQL Query

SELECT

```
`Hospital Name`,
```

```

Department,

DATEDIFF(

    STR_TO_DATE(`Discharge Date`, '%d-%m-%Y'),

    STR_TO_DATE(`Admission Date`, '%d-%m-%Y')

) AS Stay_Days

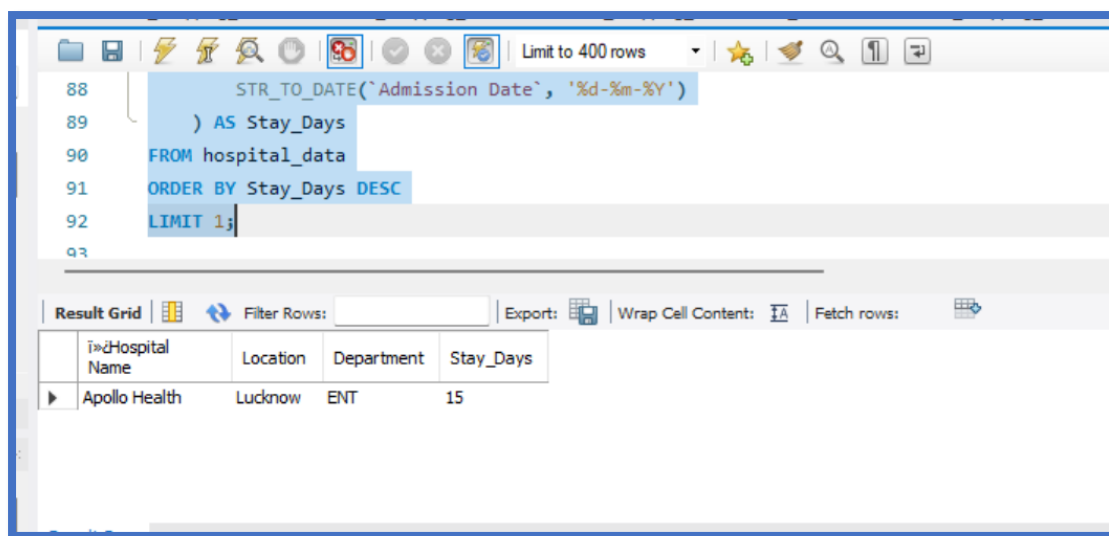
FROM hospital_data

ORDER BY Stay_Days DESC

LIMIT 1;

```

Output



The screenshot shows a SQL query editor with the following query:

```

88     STR_TO_DATE(`Admission Date`, '%d-%m-%Y')
89     ) AS Stay_Days
90 FROM hospital_data
91 ORDER BY Stay_Days DESC
92 LIMIT 1;

```

Below the query editor, the result grid displays the output of the query. The result grid has columns: Hospital Name, Location, Department, and Stay_Days. The output shows one row for Apollo Health, located in Lucknow, in the ENT department, with a stay of 15 days.

Hospital Name	Location	Department	Stay_Days
Apollo Health	Lucknow	ENT	15

Observation

The query calculates the difference between discharge and admission dates to determine the longest hospital stay.

Query 7: Total Patients Treated Per City

Objective

Calculate the total number of patients treated in each city.

SQL Query

```

SELECT

    Location,

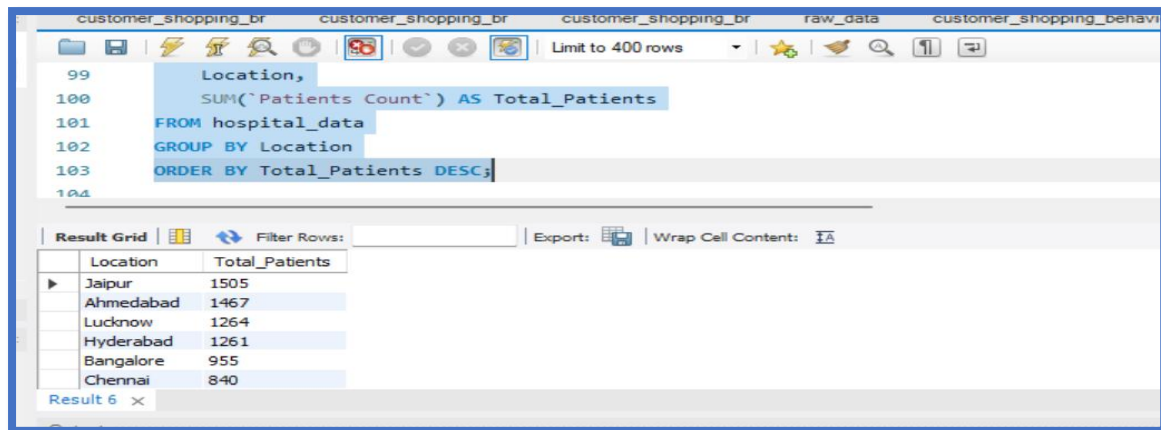
    SUM(`Patients Count`) AS Total_Patients

FROM hospital_data

```

GROUP BY Location;

Output



The screenshot shows a SQL query editor with a query window and a result grid. The query is as follows:

```
99  Location,  
100  SUM(`Patients Count`) AS Total_Patients  
101  FROM hospital_data  
102  GROUP BY Location  
103  ORDER BY Total_Patients DESC;  
104
```

The result grid displays the following data:

Location	Total_Patients
Jaipur	1505
Ahmedabad	1467
Lucknow	1264
Hyderabad	1261
Bangalore	955
Chennai	840

Observation

The query summarizes the total number of patients treated in each city.

Query 8: Average Length of Stay Per Department

Objective

Calculate the average hospital stay for each department.

SQL Query

SELECT

Department,

AVG(

DATEDIFF(

STR_TO_DATE(`Discharge Date`, '%d-%m-%Y'),

STR_TO_DATE(`Admission Date`, '%d-%m-%Y')

)

) AS Average_Stay_Days

FROM hospital_data

GROUP BY Department;

Output


```

118      2
119      ) AS Average_Stay_Days
120  FROM hospital_data
121  GROUP BY Department
122  ORDER BY Average_Stay_Days DESC;
123

```

Department	Average_Stay_Days
Neurology	9.25
Pediatrics	9.11
Urology	8.72
Oncology	8.11
ENT	8.08
Gynecology	7.67

Observation

The query calculates the average number of days patients remain admitted in each department.

Query 9: Department with Lowest Number of Patients

Objective

Identify the department with the lowest number of patients.

SQL Query

SELECT

Department,

SUM(`Patients Count`) AS Total_Patients

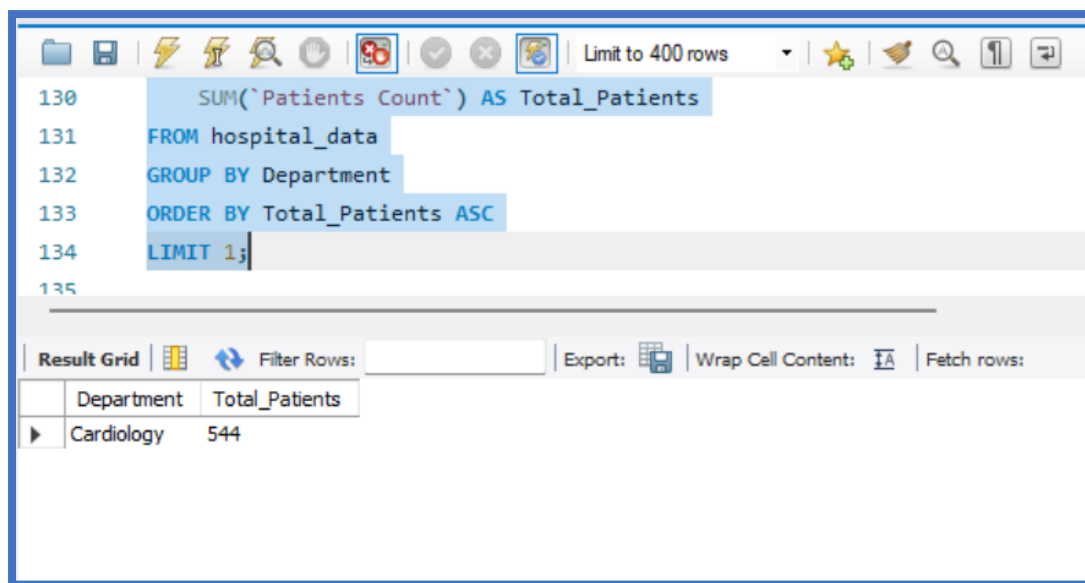
FROM hospital_data

GROUP BY Department

ORDER BY Total_Patients ASC

LIMIT 1;

Output



Observation

The query identifies the department with the fewest patients in the dataset.

Query 10: Monthly Medical Expenses Report

Objective

Calculate the total medical expenses for each month.

SQL Query

SELECT

MONTHNAME(STR_TO_DATE(`Admission Date`, '%d-%m-%Y')) AS Month,

YEAR(STR_TO_DATE(`Admission Date`, '%d-%m-%Y')) AS Year,

SUM(`Medical Expenses`) AS Total_Medical_Expenses

FROM hospital_data

GROUP BY

YEAR(STR_TO_DATE(`Admission Date`, '%d-%m-%Y')),

MONTH(STR_TO_DATE(`Admission Date`, '%d-%m-%Y')),

MONTHNAME(STR_TO_DATE(`Admission Date`, '%d-%m-%Y'))

ORDER BY

YEAR(STR_TO_DATE(`Admission Date`, '%d-%m-%Y')),

MONTH(STR_TO_DATE(`Admission Date`, '%d-%m-%Y'));

Output

```

148 MONTHNAME(STR_TO_DATE(`Admission Date`, '%d-%m-%Y'))
149 ORDER BY
150 YEAR(STR_TO_DATE(`Admission Date`, '%d-%m-%Y')),
151 MONTH(STR_TO_DATE(`Admission Date`, '%d-%m-%Y'));
152
153 -- =====

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: I A

Month	Year	Total_Medical_Expenses
January	2023	173971.54
February	2023	301722.72
March	2023	199247.42
April	2023	88995.93
May	2023	222986.71999999997
June	2023	165926.36000000002

Result 9 x

Observation

The query generates a month-wise report of the total medical expenses.

8. SQL Concepts Used

- SELECT Statement
- Aggregate Functions (SUM, AVG)
- GROUP BY
- ORDER BY
- LIMIT
- Date Functions (DATEDIFF, STR_TO_DATE, MONTHNAME, YEAR)
- Data Aggregation
- Business-Oriented SQL Analysis

9. Key Insights

After executing all SQL queries, the following insights can be derived:

- The dataset provides a comprehensive overview of hospital operations.
- Patient distribution varies across different departments.
- Some hospitals incur significantly higher medical expenses than others.
- Average hospital stay differs by department.
- City-wise patient counts help identify areas with higher healthcare demand.

- Monthly expense reports can be used to monitor healthcare spending trends.
-

10. Conclusion

This project demonstrates the practical application of SQL for healthcare data analysis. By using aggregate functions, grouping, sorting, and date functions, meaningful insights were extracted regarding patient volume, doctor availability, departmental performance, hospital stay duration, and medical expenses. The project strengthened SQL querying skills and provided hands-on experience with real-world healthcare data, making it a valuable addition to a data analyst portfolio.

References

- Dataset: Hospital_Data.csv
- Database: MySQL
- Tool: MySQL Workbench